Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z3

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z3: $\frac{Z_1 Z_3 g_m}{Z_1 g_m + 1}$

$$H(z) = \frac{Z_1 Z_3 g_m}{Z_1 g_m + 1}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 g_m}{C_1 C_3 R_3 s^2 + g_m + s\left(C_1 + C_3 R_3 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{g_m}}{C_1 + C_3 R_3 g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_3}}$
bandwidth: $\frac{C_1 + C_3 R_3 g_m}{C_1 C_3 R_3}$
K-LP: R_3
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{C_1 C_3 R_1 R_3 s^2 + R_1 g_m + s\left(C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}\sqrt{R_1 g_m + 1}}{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1}\sqrt{C_3}\sqrt{R_1}\sqrt{R_3}}$
bandwidth: $\frac{C_1 R_1 + C_3 R_1 R_3 g_m + C_3 R_3}{C_1 C_3 R_1 R_3}$
K-LP: $\frac{R_1 R_3 g_m}{R_1 g_m + 1}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s^2 \left(C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3\right) + s \left(C_1 R_1 g_m + C_1 + C_3 R_3 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_3} R_1 \sqrt{R_3} g_m \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{\frac{g_m}{R_1 g_m + 1}}}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m}$
wo: $\sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_1 C_3 R_1 R_3 g_m + C_1 C_3 R_3}} \left(C_1 R_1 g_m + C_1 + C_3 R_3 g_m\right)}{\sqrt{C_1}\sqrt{C_3} R_1 \sqrt{R_3} g_m \sqrt{\frac{g_m}{R_1 g_m + 1}} + \sqrt{C_1}\sqrt{C_3}\sqrt{R_3}\sqrt{\frac{g_m}{R_1 g_m + 1}}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_1 R_1 R_3 g_m}{C_1 R_1 g_m + C_1 + C_3 R_3 g_m}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = \left(R_1,\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + 1}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 g_m}{s \left(C_3 R_1 g_m + C_3\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1,\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{R_1 g_m + s \left(C_3 R_1 R_3 g_m + C_3 R_3\right) + 1}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{s \left(C_3 R_1 g_m + C_3\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_3 g_m}{C_1 s + g_m}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{g_m}{C_1 C_3 s^2 + C_3 g_m s}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 R_3 g_m s + g_m}{C_1 C_3 s^2 + C_3 g_m s}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_3 g_m}{C_1 R_1 s + R_1 g_m + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 g_m}{C_1 C_3 R_1 s^2 + s\left(C_3 R_1 g_m + C_3\right)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_3 R_1 R_3 g_m s + R_1 g_m}{C_1 C_3 R_1 s^2 + s\left(C_3 R_1 g_m + C_3\right)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3,\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 R_1 R_3 g_m s + R_3 g_m}{g_m + s\left(C_1 R_1 g_m + C_1\right)}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 R_1 g_m s + g_m}{C_3 g_m s + s^2\left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_3 R_1 R_3 g_m s^2 + g_m + s\left(C_1 R_1 g_m + C_3 R_3 g_m\right)}{C_3 g_m s + s^2\left(C_1 C_3 R_1 g_m + C_1 C_3\right)}$$

10 X-INVALID-WZ

11 X-PolynomialError